



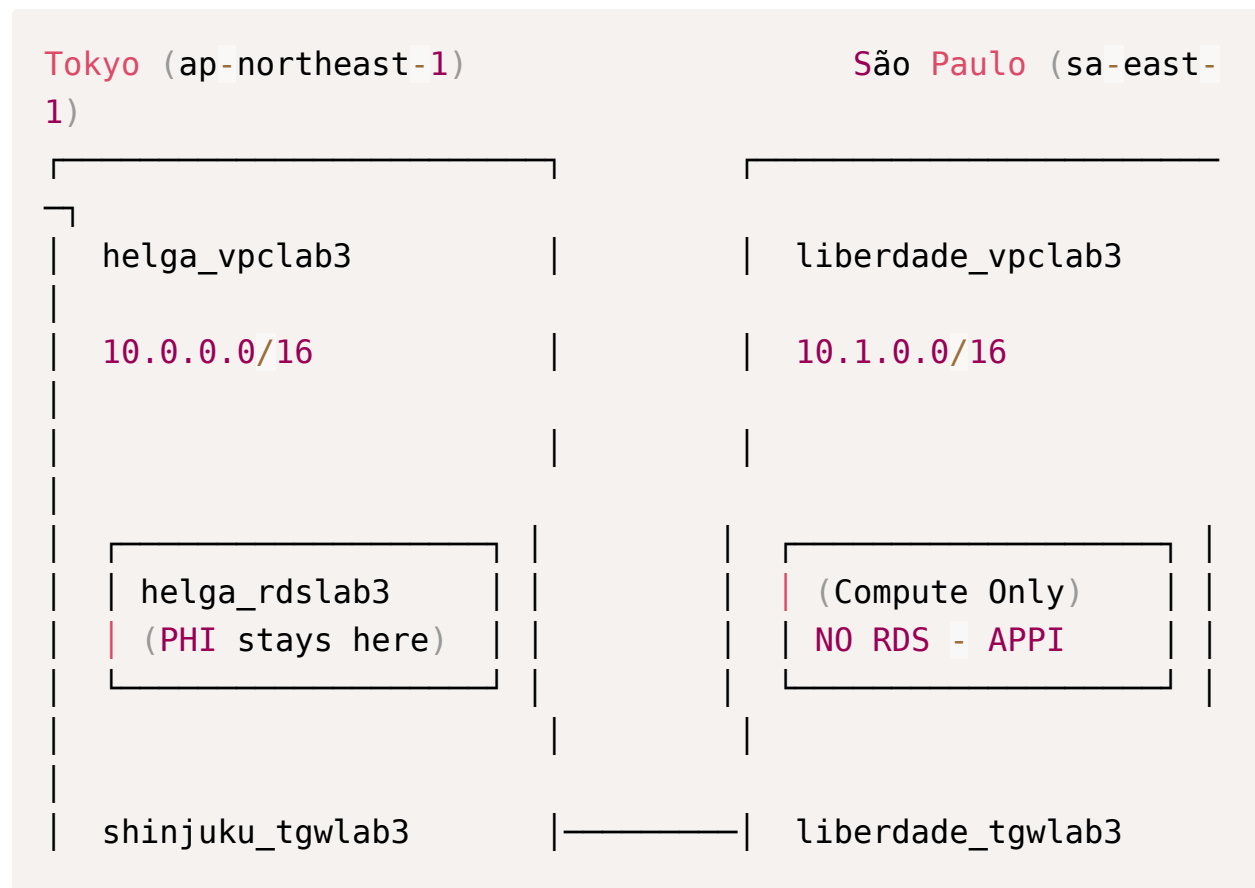
Phase 5 - Verification & Audit Pack (Sanitized)

This phase focuses on **Lab 3B deliverables**: proving APPI compliance with audit evidence.

Lab 3 Summary

What We Built

An **APPI-compliant multi-region architecture** connecting Tokyo (data authority) and São Paulo (compute spoke) via AWS Transit Gateway peering.





Phased Deployment Summary

Phase	Region	What Was Done
Phase 1	Tokyo	Created <code>shinjuku_tgwlab3</code> Transit Gateway + VPC attachment to <code>helga_vpclab3</code>
Phase 2	São Paulo	Created <code>liberdade_vpclab3</code> (VPC, subnets, NAT) + <code>liberdade_tgwlab3</code> Transit Gateway
Phase 3	Tokyo	Created TGW peering attachment (<code>shinjuku_to_liberdade_peer01</code>), VPC route to São Paulo, RDS SG rule
Phase 4	São Paulo	Accepted peering (<code>accept_tokyo_peer01</code>), added TGW route + VPC route to Tokyo
Post-Phase 4	Tokyo	Uncommented TGW route to São Paulo (required peering to be <code>available</code> first)
Phase 5	Both	Verification + added missing RDS SG ingress rule for São Paulo CIDR

Verification CLI Commands

Run these to verify Lab 3 is working correctly:

1) Verify RDS exists ONLY in Tokyo

```
# Tokyo (should have helga_rds1ab3)
echo "=== Tokyo RDS ==="
aws rds describe-db-instances --region ap-northeast-1 \
  --query "DBInstances[*].{ID:DBInstanceIdentifier,Status:DBInstanceStatus,Endpoint:Endpoint.Address}"
```

```
# São Paulo (should be EMPTY - APPI compliance)
echo "=== São Paulo RDS (should be EMPTY) ==="
aws rds describe-db-instances --region sa-east-1 \
  --query "DBInstances[*].DBInstanceIdentifier"
```

Expected: Tokyo shows `helga-rdslab3`, São Paulo shows empty list.

2) Verify TGW Peering is Active

```
# Check from Tokyo
echo "=== TGW Peering (Tokyo view) ==="
aws ec2 describe-transit-gateway-peering-attachments --region
ap-northeast-1 \
  --query "TransitGatewayPeeringAttachments[*].{State:State,R
equesterTGW:RequesterTgwInfo.TransitGatewayId,AcceptorTGW:Acc
epterTgwInfo.TransitGatewayId}"

# Check from São Paulo
echo "=== TGW Peering (São Paulo view) ==="
aws ec2 describe-transit-gateway-peering-attachments --region
sa-east-1 \
  --query "TransitGatewayPeeringAttachments[*].{State:State}"
```

Expected: State = `available` in both regions.

3) Verify Route Tables

```
# Tokyo private route table (should have route to São Paulo 1
0.1.0.0/16)
echo "=== Tokyo Routes to São Paulo ==="
aws ec2 describe-route-tables --region ap-northeast-1 \
  --filters "Name=tag:Name,Values=*helga*private*" \
  --query "RouteTables[*].Routes[?DestinationCidrBlock=='10.
1.0.0/16']"
```

```
# São Paulo private route table (should have route to Tokyo 10.0.0.0/16)
echo "=== São Paulo Routes to Tokyo ==="
aws ec2 describe-route-tables --region sa-east-1 \
  --filters "Name=tag:Name,Values=*liberdade*private*" \
  --query "RouteTables[*].Routes[?DestinationCidrBlock=='10.0.0.0/16']"
```

Expected: Both show `active` routes via TGW.

4) Verify RDS Security Group Allows São Paulo

```
echo "=== RDS SG Rules (should include 10.1.0.0/16) ==="
aws ec2 describe-security-groups --region ap-northeast-1 \
  --filters "Name=group-name,Values=*rds*" \
  --query "SecurityGroups[*].IpPermissions[?FromPort==\`3306\`].IpRanges"
```

Expected: Should include `10.1.0.0/16` (São Paulo VPC CIDR).

Lab 3 Success Criteria

Requirement	Status	Evidence
1) RDS exists ONLY in Tokyo	✓	<code>helga-rdslab3</code> in ap-northeast-1, none in sa-east-1
2) TGW Peering is active	✓	State = <code>available</code> in both regions
3) VPC routes configured correctly	✓	Tokyo → <code>10.1.0.0/16</code> , São Paulo → <code>10.0.0.0/16</code> via TGW
4) TGW routes configured correctly	✓	Both TGW route tables have cross-region entries
5) RDS SG allows São Paulo	✓	<code>helga_rds_sglab3</code> has <code>10.1.0.0/16</code> on port 3306



Lab 3 Complete when all 5 criteria are verified!

Example Verified Values

Resource	Value
Tokyo TGW ID	tgw-XXXXXXXXXXXXXXXXXX
São Paulo TGW ID	tgw-XXXXXXXXXXXXXXXXXX
Peering Attachment ID	tgw-attach-XXXXXXXXXXXXXXXXXX
Tokyo VPC CIDR	10.0.0.0/16
São Paulo VPC CIDR	10.1.0.0/16
Tokyo RDS Endpoint	<u>helga-rdslab3.XXXXXXXXXX.ap-northeast-1.rds.amazonaws.com</u>